

Interest Rates and Mortgage Delinquency

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Abstract

This paper estimates the effect of mortgage interest rates on delinquency using a fuzzy regression discontinuity design around the FICO score threshold of 620. After removing duplicates, the sample contains 221,906 loans with FICO scores between 570 and 670. Crossing the threshold lowers the initial interest rate by about 0.9–1.0 percentage points. Reduced-form estimates show borrowers above the cutoff are 2.0–2.3 percentage points less likely to become 60+ days delinquent. Two-stage least squares estimates imply that a one percentage point increase in the mortgage rate raises delinquency by roughly 2.1–3.5 percentage points. Results are robust across several checks, though density and covariate-balance tests reveal sorting around the threshold.

1 Introduction

Mortgage delinquency imposes large costs on households, lenders, and the broader economy. Understanding how interest rates affect repayment is therefore central to both mortgage market regulation and macro-prudential policy. Yet interest rates are not randomly assigned: lenders price mortgages using borrower risk and loan characteristics that may not be fully observed in the data, making a causal interpretation of raw rate-delinquency correlations difficult.

This paper exploits the FICO 620 threshold as a source of quasi-random variation in mortgage rates. Borrowers just above 620 receive lower interest rates than observably similar borrowers just below, generating a fuzzy RD design in which the cutoff instruments the rate. The resulting IV estimate is local to borrowers near the cutoff whose rates change because of the threshold (Imbens and Angrist, 1994). The setting relates to work on securitization and origination incentives (Keys et al., 2010) and on how origination channels affect delinquency (Jiang et al., 2014). We focus on a narrower pricing channel: whether the lower rate at the threshold reduces defaults.

2 Data

The final sample contains 221,906 loans after dropping 172 duplicate observations. FICO scores range from 570 to 670. Table 1 reports descriptive statistics. The mean delinquency rate is 28.5%, the mean interest rate is 7.0%, and 92.7% of loans were originated through brokers.

There are no missing values in FICO or interest rate. Missingness in age, gender, and income is present and changes discontinuously at the 620 cutoff, which is a potential threat to identification; all main regressions are therefore estimated with and without controls. Loan balance is logged to address right skew. Income is transformed as $\log(\text{income} + 1)$: approximately 31,000 observations

record zero income, reflecting co-borrowers with no independent earnings whose exclusion would introduce selection bias. LTV is left in levels—its distribution is negatively skewed and multimodal, with mass points near 20% and 80% corresponding to standard GSE origination thresholds, so a log transformation would not improve it. Two correlations deserve mention for interpretation: the interest rate is negatively correlated with LTV ($r = -0.45$), driven by GSE-conforming high-LTV loans with regulated pricing, and strongly negatively correlated with log balance ($r = -0.56$), consistent with volume discounts. The maximum VIF across all covariates is 1.58, so multicollinearity is not a concern.

Table 1: Descriptive Statistics

	Mean	Std. Dev.	p25	Median	p75
PANEL A: OUTCOME AND RUNNING VARIABLE					
Delinquency (60+ days)	0.285	0.451	0.000	0.000	1.000
FICO score	642.1	21.68	629.0	646.0	660.0
PANEL B: LOAN CHARACTERISTICS					
Interest rate	0.070	0.028	0.062	0.070	0.081
Log(loan balance)	12.08	0.793	11.63	12.21	12.66
Loan-to-value ratio	68.04	23.24	62.65	79.35	80.00
Broker originated	0.927	0.261	1.000	1.000	1.000
Hard prepayment penalty	0.091	0.287	0.000	0.000	0.000
Refinance	0.634	0.482	0.000	1.000	1.000
PANEL C: BORROWER CHARACTERISTICS					
Log(income+1)	7.449	3.179	8.087	8.622	9.048
Age	43.01	12.15	34.00	42.00	51.00
Female	0.625	0.484	0.000	1.000	1.000
White	0.639	0.480	0.000	1.000	1.000
Observations	221,906				

3 Empirical Strategy

Define $Above_i = \mathbf{1}[FICO_i \geq 620]$ and $FICO_i^c = FICO_i - 620$. The first stage, reduced form, and 2SLS second stage are:

$$rate_i = \alpha + \pi Above_i + f(FICO_i^c) + Above_i \times f(FICO_i^c) + X_i' \gamma + u_i, \quad (1)$$

$$Del_i = \alpha + \rho Above_i + f(FICO_i^c) + Above_i \times f(FICO_i^c) + X_i' \delta + \varepsilon_i, \quad (2)$$

$$Del_i = \alpha + \beta \widehat{rate}_i + f(FICO_i^c) + Above_i \times f(FICO_i^c) + X_i' \theta + \eta_i. \quad (3)$$

The baseline specification uses a ± 40 FICO-point bandwidth, a linear polynomial in centered FICO with separate slopes on each side, and standard errors clustered by area.

4 First Stage

Figure 1 shows a sharp fall in interest rates at FICO 620. Within each two-point bin the average rate is smooth on both sides of the threshold, with a visible discontinuity of approximately 0.7 percentage points at the cutoff.

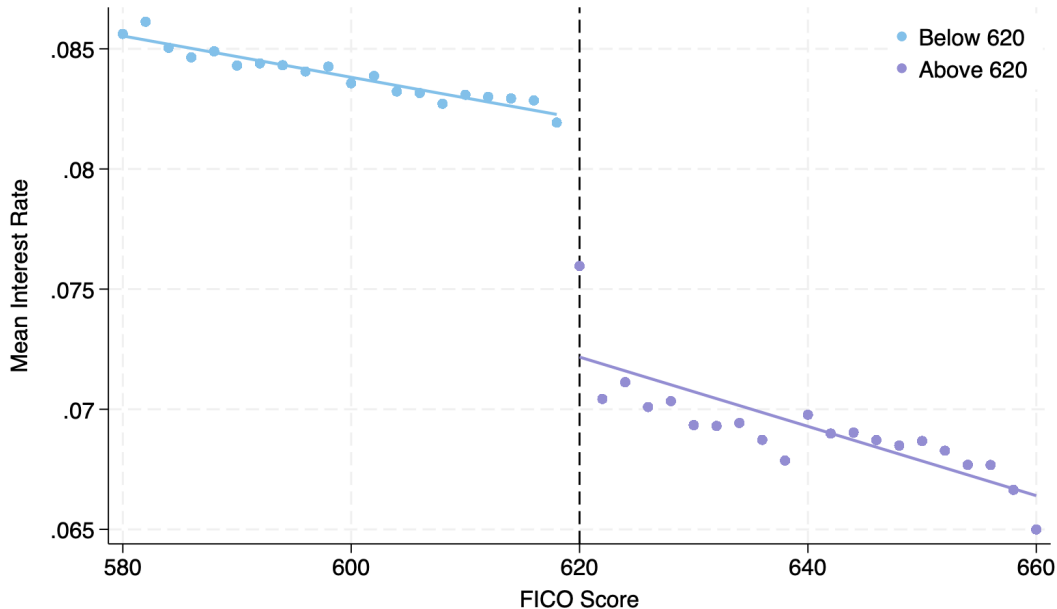


Figure 1: Interest Rate around the FICO 620 Threshold.

Table 2 confirms the result. Crossing the cutoff lowers the interest rate by 0.9–1.0 percentage points, significant at the 1% level in all specifications and stable across bandwidths and polynomial orders. We complement these estimates with local polynomial RD estimates using MSE-optimal bandwidth selection following Imbens and Kalyanaraman (2012) and Calonico et al. (2014) (Table 3). At optimal bandwidths of about five FICO points the estimated discontinuity is 0.4–0.5 percentage points, significant at the 1% level; the CER-optimal estimate is smaller and insignificant, consistent with very few observations on the left of the cutoff at that window. First-stage F -statistics in Table 4 range from 128.5 to 873.3, far above the weak-instrument benchmark of Stock and Yogo (2005).

5 Validity Checks

Density. The density test rejects continuity of the running variable ($T = 54.95$, $p < 0.001$), driven by 6,206 observations exactly at FICO 620 (McCrary, 2008; Cattaneo et al., 2020). The mass point likely reflects institutional sorting: brokers may have helped loans cross the threshold to obtain better pricing (Keys et al., 2010). If riskier borrowers were disproportionately assisted, this would suggest our estimates may be conservative. Figure A1 in the appendix shows the full density.

Table 2: First Stage: Effect of FICO ≥ 620 on Interest Rate

	± 20	± 30	± 40	± 60	± 80
PANEL A: LINEAR POLYNOMIAL					
Above 620	−0.0089*** (0.0010)	−0.0099*** (0.0010)	−0.0099*** (0.0010)	−0.0097*** (0.0010)	−0.0097*** (0.0010)
Observations	77,067	120,738	165,998	221,906	221,906
R^2	0.030	0.029	0.034	0.039	0.039
PANEL B: QUADRATIC POLYNOMIAL					
Above 620			−0.0092*** (0.0011)	−0.0095*** (0.0010)	−0.0095*** (0.0010)
Observations			165,998	221,906	221,906
R^2			0.034	0.039	0.039
PANEL C: CUBIC POLYNOMIAL					
Above 620			−0.0070*** (0.0011)	−0.0083*** (0.0011)	−0.0083*** (0.0011)
Observations			165,998	221,906	221,906
R^2			0.035	0.039	0.039

Notes: Dependent variable is the initial interest rate on the mortgage. Each cell reports the OLS coefficient on *Above 620* - an indicator equal to one if FICO ≥ 620 . All specifications include a flexible polynomial in the centered running variable (FICO $- 620$) with separate slopes on each side of the threshold. Standard errors clustered by area in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: First Stage: Local Polynomial Estimates (rdrobust)

	Uniform kernel MSE-optimal	Triangular kernel MSE-optimal	Triangular kernel CER-optimal
RD Effect	−0.0046*** (0.0010)	−0.0042*** (0.0010)	−0.0022 (0.0018)
Bandwidth h	4.80	5.19	2.80
N (left of cutoff)	2,152	2,708	1,098
N (right of cutoff)	17,577	20,471	11,932

Notes: Dependent variable is the initial interest rate. Running variable is FICO score, cutoff at 620. MSE-optimal bandwidth selected to minimize mean squared error of the RD estimator; CER-optimal bandwidth selected for coverage error rate optimality. Triangular kernel assigns linearly declining weights to observations farther from the cutoff. All estimates adjusted for mass points in the running variable. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Covariate balance. Table A1 reports RD regressions with each covariate as outcome. Gender and refinance status are smooth at the cutoff. Log balance, LTV, age, broker share, and hard prepayment penalties change discontinuously, reflecting GSE eligibility criteria that differ on each side of 620 rather than random compositional differences. All discontinuous covariates are included as controls in the main regressions; results are stable to their inclusion (Table 4).

Placebo cutoffs. A narrow ± 5 FICO-point window is used for placebo tests: over the full FICO range the interest rate declines monotonically, so any wider window around a false cutoff would mechanically pick up the smooth trend. Five of six placebo cutoffs in Table A2 are insignificant; the marginally significant coefficient at $c = 640$ may reflect a secondary pricing threshold.

Donut RDD. Excluding the 6,206 observations at FICO 620 yields a first-stage estimate of -0.0110 , compared with -0.0099 in the full sample (Table A3). The mass point does not drive the result.

Local randomization. Fisher permutation tests with 1,000 permutations (Cattaneo et al., 2015) confirm significant rate and delinquency differences at the cutoff within ± 5 and ± 10 FICO windows (Table A5), supporting the threshold effect under an alternative identifying assumption.

6 Main Results

Table 4 reports the reduced-form and 2SLS estimates. In the reduced form, crossing FICO 620 lowers delinquency by 2.0–2.3 percentage points. The coefficient is stable across control sets, indicating that the covariate imbalance documented above does not confound the result.¹ The 2SLS coefficient implies that a one percentage point increase in the mortgage rate raises delinquency by 2.1–3.5 percentage points, an increase of 7–12% relative to the mean of 28.5%. This is the local average treatment effect for compliers—borrowers near FICO 620 whose interest rate changed because of the threshold (Imbens and Angrist, 1994)—and should not be extrapolated to other populations. Bandwidth sensitivity is explored in Figure A3: the IV coefficient is positive and economically stable across all windows from ± 25 to ± 60 , and statistically significant at ± 30 and above. Results with alternative bandwidths and polynomial orders appear in Table A4.

7 Heterogeneity

Table 5 reports heterogeneity by origination channel, race, and loan purpose. The rate effect is present in both bank and broker subsamples: the reduced-form estimate is -0.0293 for bank loans and -0.0216 for broker loans, and a pooled interaction on $Above \times Broker$ is -0.046 ($p < 0.01$).

¹A probit specification yields a marginal effect of -0.030 ($p < 0.001$), slightly larger in magnitude than the LPM estimates, confirming the result is not an artefact of the linear probability model.

Table 4: Main Results: Effect of Interest Rate on Delinquency

	Reduced Form			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
Above 620	−0.0205** (0.0089)	−0.0227*** (0.0080)	−0.0231*** (0.0083)			
Interest rate				2.066*** (0.794)	3.134*** (1.094)	3.475*** (1.272)
Controls	No	Stable	All	No	Stable	All
First-stage F				128.5	672.8	873.3
Observations	165,998	165,989	149,736	165,998	165,989	149,736
R^2	0.016	0.039	0.048	0.022	0.035	0.048

Notes: Dependent variable is the delinquency indicator (=1 if 60+ days past due). Columns (1)–(3) report OLS reduced-form estimates of the effect of crossing FICO=620 on delinquency. Columns (4)–(6) report 2SLS estimates instrumenting the interest rate with *Above 620*. All specifications include a linear polynomial in centered FICO with separate slopes on each side of the threshold, within bandwidth ± 40 . “Stable controls”: log(balance), LTV, hard prepayment penalty, refinance indicator, race, and broker channel. “All controls” additionally include log(income+1), age, and gender. Probit marginal effect at the threshold (specification 2): -0.030^{***} (0.007), confirming the LPM estimates. First-stage F -statistics exceed the Stock and Yogo (2005) weak instrument threshold of 10. Standard errors clustered by area in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The bank IV estimate is larger (4.280 vs. 2.925) but less precise given the smaller bank subsample. For race, the reduced form is significant for white borrowers (-0.0271) but not for non-white borrowers, which may reflect lower precision or the finding that non-white borrowers face higher rates conditional on FICO regardless of the threshold (Jiang et al., 2014). By loan purpose, the effect is concentrated among refinance loans (reduced form -0.0290 , IV 2.712, both significant at the 1% level); purchase loans show no significant effect, consistent with a weaker first stage in that subsample and with refinance borrowers being inherently more sensitive to rate changes.

Table 5: Heterogeneity Analysis

	Broker channel		Race		Loan purpose	
	Bank	Broker	Non-White	White	Purchase	Refinance
PANEL A: REDUCED FORM						
Above 620	−0.0293* (0.0173)	−0.0216** (0.0091)	−0.0181 (0.0127)	−0.0271*** (0.0084)	−0.0109 (0.0160)	−0.0290*** (0.0086)
PANEL B: 2SLS						
Interest rate	4.280* (2.593)	2.925** (1.203)	3.241 (2.281)	3.280*** (0.996)	12.929 (17.808)	2.712*** (0.816)
Observations	12,295	153,694	60,213	105,776	59,770	106,219

Notes: Each column reports estimates from a separate split-sample regression. Panel A: reduced-form OLS. Panel B: 2SLS instrumenting interest rate with *Above 620*. All specifications use bandwidth ± 40 , linear polynomial with separate slopes, and stable controls appropriate for each subsample. The differential effect by broker channel is confirmed by a formal interaction test: the coefficient on *Above*×*Broker* in the pooled reduced-form regression is -0.046^{***} (0.012). Imprecise IV estimates for Purchase loans reflect a weak first stage in that subsample. Standard errors clustered by area in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

8 Conclusion

The FICO 620 threshold generates a sharp fall in mortgage rates. Borrowers just above the cutoff are 2.0–2.3 percentage points less likely to become delinquent, and the IV estimates imply that a one percentage point increase in the rate raises delinquency by roughly 2.1–3.5 percentage points. The result is robust to bandwidth choice, polynomial order, donut exclusion, placebo cut-offs, and local-randomization inference. Sorting at the threshold limits the clean-RD interpretation and suggests our estimates are conservative. The findings point to a pricing channel alongside the screening channel of Keys et al. (2010): credit-score thresholds affect repayment not only through origination incentives but also through the direct affordability of the mortgage.

References

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A Additional Figures and Tables

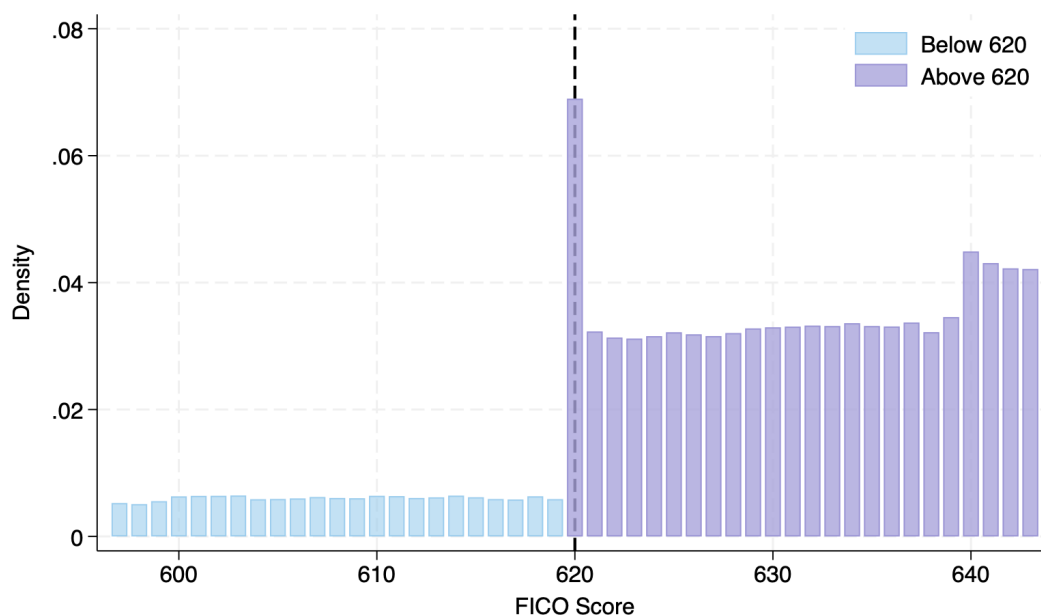


Figure A1: Density of FICO Scores around the 620 Threshold ($T = 54.95$, $p < 0.001$).

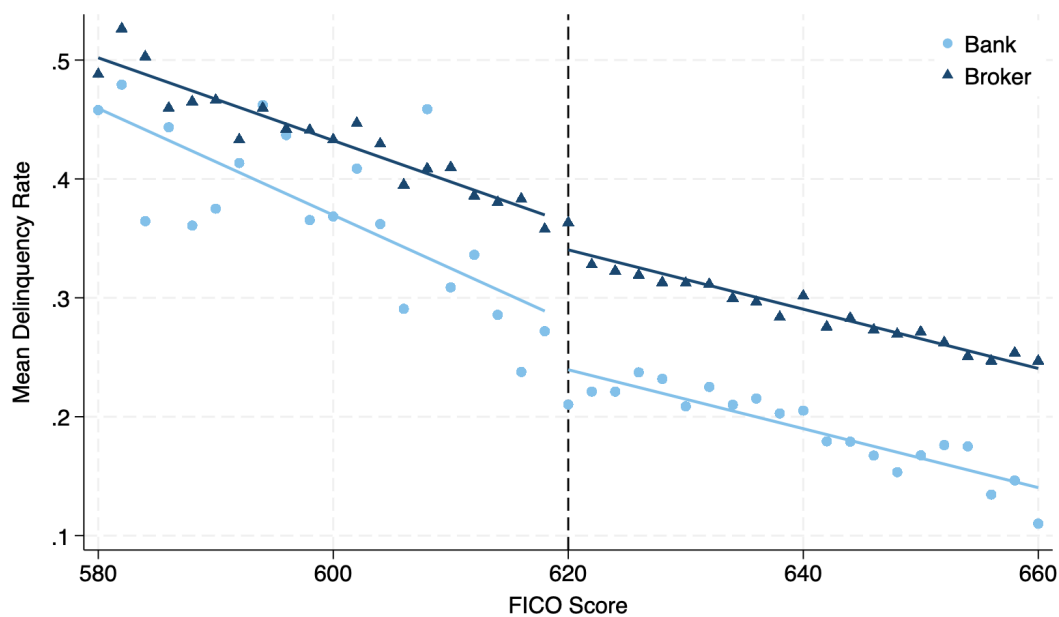


Figure A2: Delinquency around FICO 620 by Origination Channel.

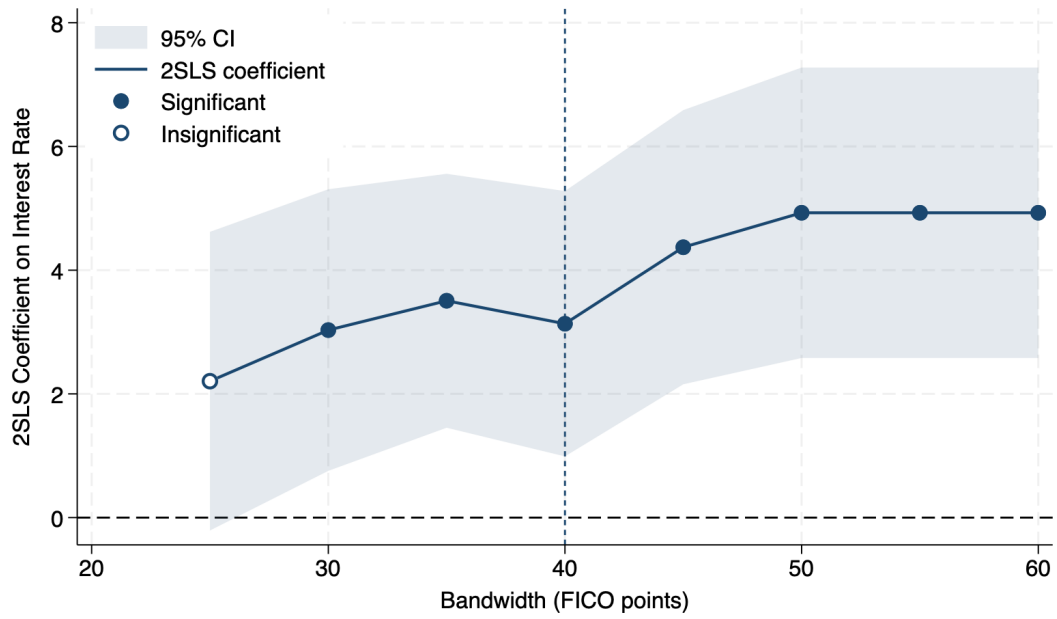


Figure A3: Bandwidth Sensitivity of the 2SLS Estimate.

Table A1: Covariate Balance at FICO = 620 Threshold

	log(balance)	log(income+1)	LTV	Age	Broker	Gender	Hard	Refinance	White
Above 620	0.2018*** (0.0265)	-1.2229*** (0.0912)	-2.4935*** (0.4481)	-1.0139*** (0.2813)	0.0432*** (0.0053)	0.0033 (0.0075)	0.0660*** (0.0042)	0.0075 (0.0165)	-0.0185* (0.0097)
Observations	165,998	158,591	165,989	156,710	165,998	158,615	165,998	165,998	165,998

Notes: Each column reports the OLS coefficient on *Above 620* from a separate regression of the covariate on the threshold indicator and a linear polynomial in centered FICO with separate slopes on each side of the threshold, within bandwidth ± 40 . Gender and Refinance show no significant discontinuity, supporting the validity of the RDD design. Standard errors clustered by area in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Placebo Cutoff Tests

	c=590	c=600	c=610	c=630	c=640	c=650
Placebo Above	-0.0005 (0.0009)	-0.0007 (0.0011)	0.0005 (0.0008)	-0.0004 (0.0005)	0.0021*** (0.0007)	0.0005 (0.0005)
Observations	5,482	5,763	6,102	32,350	38,461	42,790

Notes: Each column reports the coefficient on a placebo threshold indicator from a separate OLS regression of the interest rate on the placebo indicator and a linear polynomial in the recentered FICO score with separate slopes, within bandwidth ± 5 FICO points of the placebo cutoff. Five of six placebo cutoffs yield statistically insignificant coefficients, supporting the specificity of the FICO=620 discontinuity. The significant coefficient at $c = 640$ may reflect a secondary pricing threshold unrelated to our identification strategy. Standard errors clustered by area in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Donut RDD Robustness Check

	Full sample	Donut (excl. FICO=620)
Above 620	−0.0099*** (0.0010)	−0.0110*** (0.0010)
Observations	165,998	159,792

Notes: Dependent variable is the initial interest rate. Both specifications use bandwidth ± 40 and a linear polynomial with separate slopes on each side of the threshold. The donut specification excludes the 6,206 observations with FICO=620 to test whether results are driven by the mass point at the threshold. The effect is slightly larger in the donut sample, confirming that the mass point does not inflate our estimates. Standard errors clustered by area in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Robustness: Varying Bandwidth and Polynomial Order

	Reduced Form		2SLS	
	Linear, ± 60	Quadratic, ± 80	Linear, ± 60	Quadratic, ± 80
Above 620	−0.0310*** (0.0073)	−0.0136 (0.0099)		
Interest rate			4.928*** (1.198)	1.755 (1.236)
Observations	221,893	221,893	221,893	221,893
R^2	0.040	0.040	0.010	0.042

Notes: All specifications include stable controls (log(balance), LTV, hard prepayment penalty, refinance, race, broker channel) and separate polynomial slopes on each side of the threshold. The quadratic specification with ± 80 bandwidth yields imprecise estimates due to the combination of a wide window and a flexible polynomial, consistent with a bias-variance tradeoff. Standard errors clustered by area in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Local Randomization Test

Outcome	Window ± 5		Window ± 10	
	Diff. in means	p -value	Diff. in means	p -value
Delinquency	−0.026	0.004	−0.050	< 0.001
Interest rate	−0.009	< 0.001	−0.011	< 0.001

Notes: Within narrow windows around the FICO=620 threshold, the running variable is assumed to be locally randomized. The test compares means of the outcome on each side of the cutoff using Fisher randomization inference with 1,000 permutations. Significant differences in both delinquency and interest rate confirm the threshold effect under this alternative identification assumption.